
Neo Mohsenvand

Applied AI Engineer | Agent Systems & Enterprise Deployment

Irving, TX | mmv@mit.edu | +1-857-500-0980

neovand.github.io | github.com/NeoVand | linkedin.com/in/mohsenvand

AI engineer building enterprise agent systems, evaluation infrastructure, and local AI tools. Experience spans financial analysis, invoice processing, voice-based stakeholder interviews, and browser ML applications. MIT PhD with research in self-supervised learning and human-AI interaction.

TECHNICAL SKILLS

Applications | Python, TypeScript, SvelteKit, React, Azure, Snowflake

Agents and evaluation | LangGraph, MLflow, DeepEval; retrieval, real-time voice, human-in-the-loop workflows

ML and local inference | PyTorch, JAX, Ollama, Transformers.js, WebGPU

EXPERIENCE

Caterpillar Inc. | April 2024 - Present

Chief AI Architect, Global Finance | January 2026 - Present

- Co-developed an invoice-processing agent connecting mailbox ingestion, PDF and scanned-document OCR, Snowflake lookups, and human exception review. Reduced the share of invoices requiring **human review from 100% to approximately 10%** in the workflow.
- Developed **Tempo**, an enterprise investigation platform that identifies stakeholders through the org chart, schedules meetings, conducts OpenAI Realtime voice interviews, and populates structured forms. Available to hundreds of internal users and actively used for idea intake, requirements gathering, and conflict resolution.
- Developed **Aperture**, an agentic image-generation system that supports slide-deck creation and other visual work from short prompts and hundreds of reference inputs.
- Created **10+ complete example applications** for an agentic-AI workshop now in its fifth iteration, including agentic RAG, talking dashboards, and voice-based data-science assistants; teach engineers and senior leaders across the company.

Senior Principal Data Scientist, Global Finance | May 2025 - January 2026

- Developed and deployed **Penman** using SvelteKit and Azure to analyze earnings-call transcripts and regulatory filings for finance leaders. Built an **MLflow/DeepEval evaluation system** and coordinated benchmark development across the organization.
- Built an AI-idea prioritization agent using a RICE-derived methodology, company-specific impact measures, and human review.
- Designed an adversarial LLM pipeline to test simulated trading-agent interpretations of draft disclosures and identify potentially unintended readings.

Senior Engineering Fellow, Artificial Intelligence | April 2024 - May 2025

- Founding technical hire of the CAT Tech AI team; led four engineers delivering **12+ applications serving 8,000+ technical users** and generating **\$8.7M in measured business value** through engineering-analysis automation.
- Created **Magic Table** for spreadsheet and database workflows over million-row datasets, combining formulas with local-LLM prompts that generate rows and columns and trigger iterative operations; developed versions using Ollama.
- Established an AI curriculum and reading group across 10+ teams, training 300+ engineers.

EARLIER EXPERIENCE AND RESEARCH

BrainCo | Head of Research | November 2021 - March 2024

- Led 11 researchers developing wearable AI and brain-computer interfaces; named inventor on five provisional patent applications.
- Led development of a conditional-GAN system reconstructing breathing patterns from 25 Hz wearable PPG, with a one-second breathing-curve feedback delay and respiratory-rate error of 1.47 breaths/min MAE across ten test subjects.
- Led soft-speech interface research comparing contact and MEMS microphones. In an eleven-person study, contact microphones reduced soft-speech word error rate from 99-100% to 35% in the tested music/noise conditions.

MIT Media Lab | Postdoctoral Research Scientist, part-time | November 2021 - June 2022

- Conducted research on multimodal memory systems and physiological data.

Apple | Machine Learning Research Intern | June - September 2021

- Developed parameter-efficient transformers for biosignal event detection and methods for inspecting attention heads and activation patterns in continuous time-series models.

Cortico | Information Design and Data Visualization Intern | Summer 2017

- Built a browser-based WebGL visualization engine for dynamic social-network graphs.

University of British Columbia | Data Scientist and Researcher | September 2013 - August 2014

- Developed a Python library for retrieving, composing, and visualizing Allen Brain Atlas connectivity data in 3D.

SELECTED PUBLIC ENGINEERING AND EDUCATION

harnessXray | Agent observability lab using LangGraph and Deep Agents, with streamed request capture, record/replay, persistent checkpoints, and source-validation tests.

Voicebook | Document-to-speech application with browser ONNX inference, WebGPU/WASM execution, local storage, optional cloud-model integrations, and end-to-end browser tests.

Tissue | Browser transformer training and interpretability lab with activation capture, neuron ablations, checkpoints, and numerical tests.

LangX, jaxverse, Embedding Playground | Built interactive AI education tools with executable agent workflows, live browser model training, and hands-on retrieval and embedding experiments.

EDUCATION

MIT | PhD, Media Arts and Sciences. Completed 2021; MIT degree record: February 2022.

Doctoral research: first author of SeqCLR (2020), adapting contrastive learning to EEG and evaluating emotion recognition, sleep staging, and abnormal-EEG detection across three datasets.

University of Oxford | MSc, Mathematical Modelling and Scientific Computing | 2013

Tehran Polytechnic | BSc, Biomedical Engineering; BSc, Electrical Engineering | 2011

SELECTED PUBLICATIONS

Contrastive Representation Learning for Electroencephalogram Classification. First author. ML4H NeurIPS Workshop, PMLR 136 (2020).

Soft Speech, Loud World: Bone Conduction Microphones Enhance Voice Assistant Interaction. Coauthor. IEEE ICCE (2024).

Generating Breathing Patterns in Real-Time: Low-Latency Respiratory Phase Tracking From 25 Hz PPG. Corresponding author. HealthyIoT / HealthWear, Springer LNICST 530 (2024).